

- 68** The table shows details of genetic characteristics of three species of organisms and the genetic crosses carried out with them.

<i>cross</i>	<i>organism</i>	<i>dominant characteristic</i>	<i>recessive characteristic</i>	<i>description of parents in cross</i>	
V	cattle	have horns	hornless	heterozygous horned	× hornless
W	fruit flies	grey-bodied	ebony-bodied	heterozygous grey-bodied	× heterozygous grey-bodied
Y	pea plants	round seeds	wrinkled seeds	wrinkled seeds	× homozygous round seeds

All alleles exhibit simple dominant/recessive relationships.

Which of the following statements is/are correct?

- 1** If two of the offspring from cross V are bred together, three different crosses could result, with three different probabilities of phenotypes in the next generation: 100% hornless, 50% hornless, or 25% hornless.
 - 2** The genotypes of the offspring with the dominant characteristic from cross W will be identifiable through their phenotypes.
 - 3** A farmer planted seeds collected from the offspring of cross Y. If these seeds grew and produced flowers which were then cross-pollinated with each other, the resulting seeds would yield a crop containing 50% homozygous plants.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3